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11 What Does the Head Direction Cell System Actually Do?

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The firing of head direction (HD) cells is one of the clearest, most robust, neural signals observed in the mammalian brain. As discussed in the preceding chapters, we know much about how the activity of these cells is controlled by cues in the animal's environment, and we are beginning to understand the critical circuitry underlying the processing of this signal. However, an obvious question with regard to this signal remains: what is its purpose? This chapter provides a review of the data on the relationship between head direction cells and behavior. (The interested reader may wish to consult Muir and Taube (2002) for a review that also addresses how place cells might guide behavior.) This chapter will be concerned primarily with electrophysiological studies. (A review of the behavioral effects of lesions to the HD cell system can be found in chapter 13 by Aggleton.) As the findings described in this chapter will show, although initial studies have demonstrated a clear relation between HD cells and behavior, recent work has shown that this link does not apply in all spatial tasks.

Before considering individual findings, we would like to note that the studies linking HD cells and behavior, and indeed hippocampal place cells and behavior, are correlational. That is, we, as experimenters, observe changes in the HD or place cell representation of an environment, and look for corresponding changes in the animal's behavior. The assumption underlying this approach is that such a correspondence is consistent with (but certainly does not prove) the notion that the HD cell system guides specific behaviors. On the other hand, a negative effect—a lack of correlation—provides strong evidence against the participation of this system in a given behavior. More direct means of testing the system by, for example, electrically stimulating individual neurons, may seem like an attractive alternative. However, given that HD cells are found in a variety of brain regions, a negative effect—a lack of change in behavior following stimulation of the HD cells in a particular brain region—would be quite difficult to interpret.

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Of Rats and Mazes: Correlation between Head Direction Cells and Behavior

Since the first maze studies by Small (1901), many psychological studies have capitalized on the exceptional ability of rats to use spatial information to solve spatial tasks. Even in the early studies of Watson (1907) it was recognized that "the so-called 'sense of position' ('sense of direction') in this animal is extremely well marked" (p. 85). Lashley (1929) suggested that in learning to run a maze, rats developed "some central organization by which the sense of general direction can be maintained in spite of great variations of posture and of specific direction in running" (p. 138). Later work also provided evidence that rats use "spatial direction cues" when alternating on a T-maze (Douglas, 1966, p. 179). These behavioral observations, of course, were made well in advance of the discovery of head direction (HD) cells in the 1980s.

Does the HD cell system underlie a spatial direction sense? Surprisingly, only a few studies have addressed this question. The purpose of this chapter is to examine these data, first considering supporting evidence, then considering refuting evidence, and then attempting to reconcile the two. We will conclude with a brief discussion of related findings from experiments using disorientation.

An initial attempt to relate HD cells and spatial behavior is found in the study by Mizumori and Williams (1993), which describes HD cells in the lateral dorsal thalamus. In two animals, they observed an increase in the directionality of HD cells that correlated with a decrease in the number of mistakes made in learning a radial arm maze task.

In a more explicit study of the relationship between HD cells and behavior, Dudchenko and Taube (1997) trained rats to select a single reinforced maze arm on a plus- or radial-arm maze (comparable to the 1987 study by O'Keefe and Speakman of hippocampal place cells). The maze was curtained off from the remainder of the recording environment, and a white curtain occupying 48° of the black-curtained enclosure served as a salient visual landmark in the environment. The question these authors asked was, can a rat's behavior on the maze be predicted by the behavior of its HD cells?

To test this, once rats had learned to select readily the maze arm that contained reinforcement, the white "cue" curtain was rotated by 90° or 180° (in the absence of the rat). If this cue anchored the head direction cell system, then the preferred firing direction of individual HD cells would be expected to rotate by 90° or 180°. If the cue curtain also exerted stimulus control over the rats' spatial behavior, then the rats would be expected to choose an arm with a 90° or 180° orientation relative to the maze arm chosen prior to curtain rotation.

An example of the results from these manipulations can be seen in figure 11.1. Following training, the rat in this example reliably selected maze arm 1 on six consecutive trials, and then chose the opposite (180°) arm following the 180° rotation of the cue curtain (figure 11.1A). (On two trials the rats first entered the arm that they faced when placed on the center of the maze, before entering arm 5 on their second choice.) The firing of a HD

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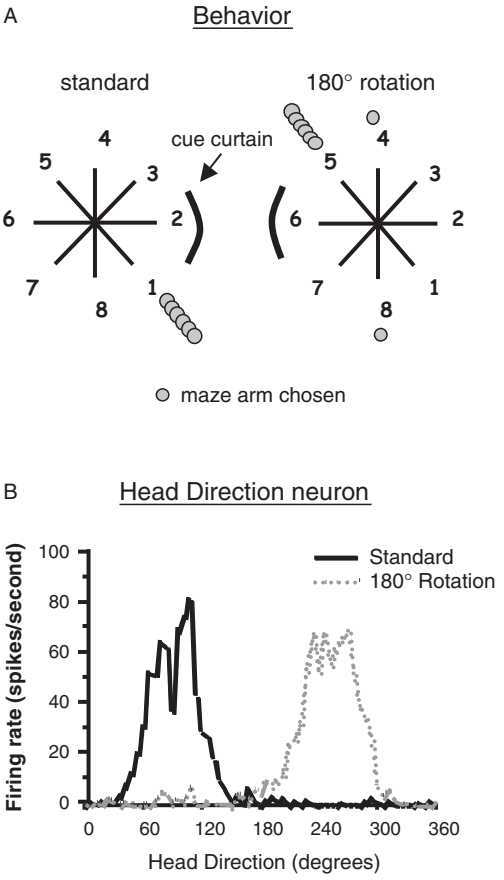


Figure 11.1
Correlation between head direction cell activity and behavior on the radial arm maze. (A) Maze arm choices (shown as dots) of a rat trained to select arm 1. Following a 180° rotation of the cue curtain, this rat selected maze arm 5, the arm with the same spatial relationship to the cue curtain as the trained arm; (B) The firing direction of an HD cell recorded during these trial also shifted by 180° during the curtain rotation trials. (From Dudchenko and Taube, 1997 with permission. Copyright American Psychological Association.)

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cell during these maze choices is shown in figure 11.1B. As is evident in this figure, 180° rotation of the cue curtain resulted in a corresponding rotation of the direction in which this cell fired. Thus, the shift in the HD cell preferred direction correlated with the shift in the rat's spatial behavior. Over both 90° and 180° rotations this correlation was robust ($r = .816$; $p < .01$; cells were recorded in either the anterior dorsal thalamus or the postsubiculum of eight rats).

Although it is possible that these changes in HD cell preferred direction and spatial behavior were independent, the observed correlation between the two is consistent with the view that HD cells provide a directional framework that can be used to guide the rats' spatial behavior. Subsequent studies, however, have suggested that this conclusion may not apply to all spatial tasks.

A Lack of Correlation between Head Direction Cells and Spatial Behavior

An important study by Golob et al. (2001) has suggested that the link between HD cells and spatial behavior is *not* universal. The authors used a spatial reference memory task where they first trained water-deprived rats to run to one corner of a square, gray box to receive a water reward. The rats were placed in the box at different entry points for each trial. Different rats were trained to run to different corners, but for each rat the correct (reinforced) corner of the box maintained a constant spatial relationship with a white cue card mounted on one wall of the box. After the rats had learned to readily run to the reinforced corner of the box (77% correct responses), they were given a probe session in a rectangular box (see figure 11.2). This novel rectangular box was similar to the square, gray box in that it also contained a white cue card, and the rat was also reinforced for choosing the corner that possessed the same spatial relationship with the cue card.

The idea behind the experiment was as follows: previous work had shown that moving a rat from a square to a rectangular apparatus causes HD cells to adopt a different pre-

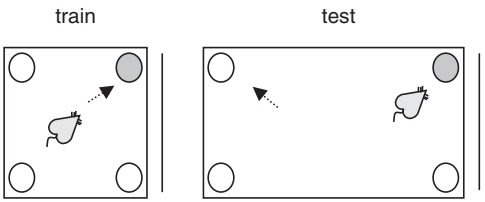


Figure 11.2
Schematic of the spatial task used by Golob et al. (2001). Rats were trained to find a reward in one corner of a square box with a cue card (train, left figure). Upon being placed in a rectangle with a similar cue card (test, right figure), the rats selected the corner of the apparatus with the same spatial relationship to the cue card, even though their HD cells shifted to a different firing direction (arrows). Vertical bars in both figures indicate position of cue card from an overhead view. The shaded circle indicates the corner in which reward was available.

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ferred direction (Taube et al., 1990). The question posed by Golob et al. was the following: If the preferred firing direction of HD cells shifts in the rectangle, will the corner that the rat chooses also shift a similar amount?

The surprising answer was "no." In 12 of 13 (92%) of the rectangle sessions, the HD cell's preferred firing direction shifted relative to the square. Most of these shifts were of at least 90°. The behavior of the rats, however, did not change. Rats selected the correct corner as accurately in the rectangle (78%) as they had in the square (77%). Moreover, the rats began choosing the correct corner in their first trials in the rectangle, suggesting that their spatial learning had generalized from the square.

A second experiment using a spatial working memory task produced a similar result. Golob et al. trained water-restricted rats on a delayed matching-to-position task in a rectangular apparatus. As before, the rectangle contained a white cue card and the animal's task on a given trial was to remember which corner was rewarded. Previous work with this task had shown that disoriented rats tend to use the shape of the environment, as opposed to visual landmarks, to guide their responses (Cheng, 1986; Margules and Gallistel, 1988). Thus, rats (and young children; Hermer and Spelke, 1994) tend not to distinguish between the correct corner of the rectangle and the corner located symmetrically opposite to this. That is to say, if a rat has been trained to go to a specific corner of a rectangular apparatus, it will use the shape of the apparatus, as opposed to a cue card, to guide its behavior.

In the Golob et al. study, after the sample phase of a trial, the rat was removed from the rectangle for a short delay. The rat was then replaced in the rectangle for the choice phase of the trial, in which its task was to remember which corner of the rectangle had provided a water reward. This task differed from the square → rectangle task previously described in that different corners were rewarded in different trial blocks. The authors found that the spatial relationship between the corner chosen by the rat and the firing direction of its HD cells was maintained on the choice phase in only 23 of 70 (33%) of the trials in which an HD cell was recorded. Thus, the rat's choice could not be predicted on the basis of its HD cells.

Interim Summary

The experiments of Golob et al. suggest that knowing the direction or change in preferred direction of individual HD cells does not allow one to predict a rat's spatial behavior in the square or rectangular chamber. Similar results have also been reported with hippocampal place cells (Jeffery et al., 2003). How can these results be reconciled with the strong correlation between HD cell-firing direction changes and maze arm choices observed by Dudchenko and Taube (1997)?

A necessary conclusion is that rats do not always make their choices using the same information that HD cells use to establish their firing directions. In the Dudchenko and Taube study, the radial maze did not possess a polarizing geometry. That is, there was

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nothing about the shape of the maze that provided information about which direction the rat was facing. Thus, the rats in this study may have been more inclined to use the only polarizing cue available, the white cue curtain, to anchor the HD cell system and to guide spatial responses. In contrast, in the square and rectangular apparatuses used by Golob et al., the geometry of the apparatus provided polarizing information in addition to the cue card. Thus, the HD cell system could at times anchor itself to one of the corners of the square or rectangle, instead of the cue card, even though the animal's behavior was guided by the other polarizing cue. Support for this view comes from the observations of Golob et al. that in the square apparatus (1) the preferred firing directions of HD cells occasionally rotated on baseline sessions even though the cue card was not moved, (2) the preferred firing direction appeared to rotate less reliably with rotation of the cue card, compared to similar experiments they conducted in a cylindrical chamber, and (3) shifts of the preferred firing directions were usually in multiples of 90°. Similarly, in the rectangle, HD cells maintained the same preferred direction across sample and choice trials only 56% of the time, even though the cue card was in the same place. These observations suggest that the cue card exerted much less stimulus control over the HD cell system in the square and rectangle than is typically observed in nongeometrically polarized environments such as the cylinder.[1]

The results of Golob et al. demonstrate that the HD cell system and the rats' spatial behavior are not tightly linked in all instances. The outstanding question then is, under what conditions, if any, is the HD cell system used? More recent work has attempted to address this issue, using different types of spatial tasks.

Do HD Cells Underlie Different Types of Spatial Behavior?

The Dudchenko and Taube (1997) and Golob et al. (2001) experiments used landmark navigation tasks that required associative learning, and found that HD cell orientation and behavior are not always coupled. However, in the absence of visual landmarks, rodents can make a spontaneous, direct return to a homesite following an excursion from that site (Etienne et al., 1988; Whishaw et al., 2001). It is thought that this ability is based on *path integration*—the integration of self-motion information provided by vestibular, kinesthetic, motor efference, and visual flow senses. There is also evidence that rats can maintain a sense of direction when they cross between different environments to solve a spatial task (Douglas, 1966; Dudchenko and Davidson, 2002). Finally, rats may be able to construct a cognitive map of an environment to take a "shortcut" to a goal location (Tolman et al., 1946). If such tasks depend on a stable direction sense then we might expect to find a tighter coupling between HD cell orientation and behavior. Next, we describe three experiments designed to test the relationship between the HD cell system and these types of spatial cognition.

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Path Integration in the Circular Maze

One spatial task that appears to rely on path integration is the "food-carrying task" designed by Whishaw and Tomie (1997). The task is a modification of the original circular holeboard maze task of Barnes (1979) that required animals to escape into a "safe" escape hole on a circular platform that contained about 18 other blocked holes spaced evenly around its periphery. The task as originally designed by Barnes involved aversive avoidance, where rats tried to avoid bright lights by escaping down into the hole. Whishaw and colleagues modified this task by using appetitive reinforcement (comparable to experimental designs of Etienne et al., 1988). Their apparatus was similar to the circle-holeboard maze, but the rat started from a nesting box beneath one of the holes. It climbed through the hole onto the platform and explored until it located a large-sized food pellet placed in one of 10 to 15 small cups positioned randomly over the platform. The pellet was too large to be consumed quickly; the rats preferred to carry it back to the nest and consume it there. The task was frequently run in the dark or with a blindfold on the rats so that path integration was the only strategy the animals could use to keep track of their orientation and return to the nest directly. Thus, it is a good task for assessing whether HD cells underlie non-landmark-based navigation. Indeed, a recent study showed that animals with lesions of the anterior dorsal thalamic nucleus (ADN) had poor heading angles on their return to the nest in the blindfold version of this task, but not in the visual version when landmarks were available, suggesting that the ADN is critically involved in path integration (Frohardt et al., 2001).

In a preliminary study, Frohardt et al. (2002) modified the food-carrying task to make it compatible with single-unit recording. The 6-foot diameter platform was positioned in the center of an 8-foot diameter, floor-to-ceiling black curtain. A white curtain that subtended approximately 30° was attached to the inside of the black curtain along one side and served as the most prominent polarizing visual cue during light conditions. Instead of holes located around the periphery, there were eight evenly spaced doorways, with each doorway covered by a black curtain (figure 11.3A). Only one doorway led to the nesting box, while the other seven doorways were blocked with pieces of wood. Once the rats were proficient at the food-carrying task in the light, they were habituated to wearing blindfolds and further trained to perform the task without visual cues. Following training, proficiency was evaluated over the course of several days in both the visual and blindfolded versions of the task. Each rat then received an electrode implantation aimed at the ADN. Once the rats recovered from surgery, they were screened daily for HD cells in a standard cylinder that rested on top of the food-carrying apparatus in the same room. After an HD cell was identified, it was recorded first in the cylinder, then during four trials in the food-carrying task, and again back in the cylinder. Four types of sessions were conducted for the food-carrying task: (1) *standard*—four trials with the refuge in the original training position; (2) *blindfold*—identical to the standard session except the rat wore a blindfold, and an additional cylinder baseline session was recorded before the food-carrying session;

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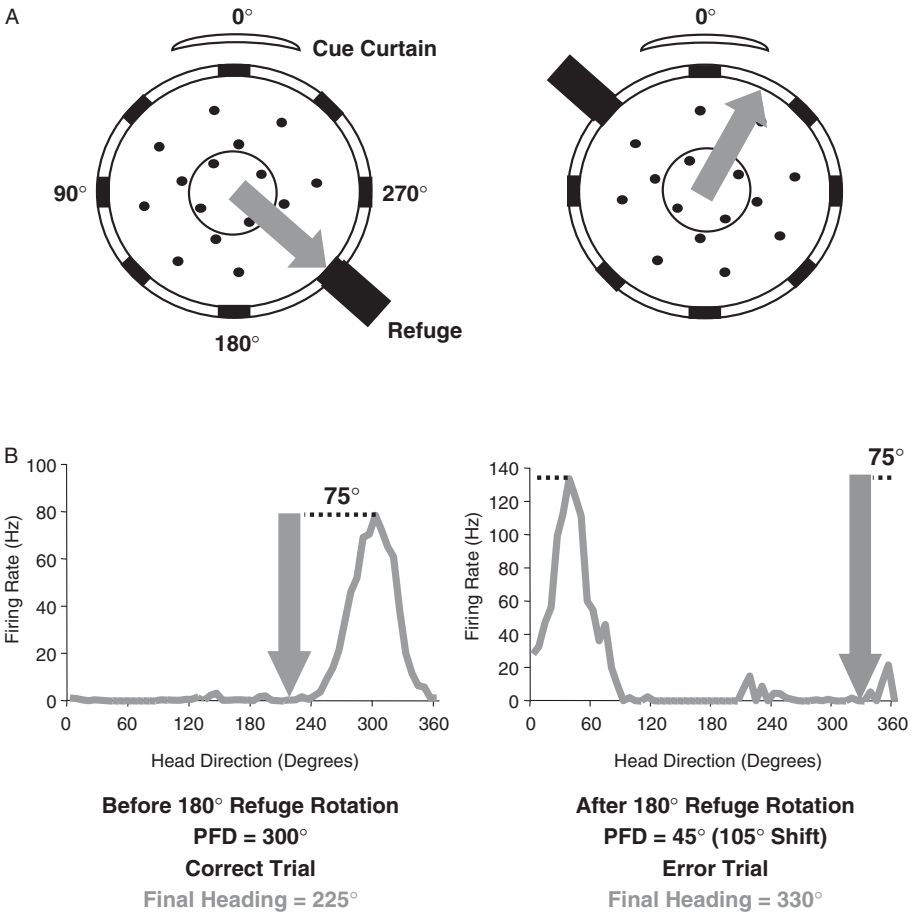


Figure 11.3
Consistency between the preferred firing direction of a head direction cell and the rat's behavioral choice after refuge rotation (A) Schematic of the food-carrying apparatus and the cue curtain located at 0°. Left: The refuge is initially located at 225° during the original training sessions. Right: The refuge and rat are rotated 180° prior to the start of the rotation session. The arrows indicate the rat's final heading during a correct trial (left) and an incorrect trial (right). (B) Corresponding firing rate vs. head direction plots from a head direction cell recorded during the sessions shown in *A*. Left graph: Correct trial. The final path heading of the rat (arrow) and the HD cell's preferred firing direction are 75° apart. Right graph: Error trial following 180° rotation of the refuge. The preferred firing direction of the HD cell shifted 105° CCW instead of 180°. Note, however, that the final path heading of the rat (arrow) and the HD cell's preferred firing direction remained 75° apart. PFD, preferred direction.

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(3) *dark*—identical to the blindfold session except darkness (room lights out) was substituted for the blindfold, and (4) *rotation*—identical to standard, except the refuge was moved to a position either 90° or 180° from the original training position, relative to the room and curtain. The purpose of the blindfold and dark sessions was to determine the rat's performance and how HD cells respond in a task that must be solved using only path integration. In contrast, the purpose of the rotation sessions was to determine under lit conditions which cues were controlling the cell's preferred firing direction and guiding the animal's behavior—allothetic cues (i.e., the cue curtain) or idiothetic/other intramaze cues. Rotations were conducted with the rat outside the apparatus, between the baseline cylinder session and the first trial of the food-carrying task session. For each rotation session there were four trials in which we monitored the cell(s) as the rat searched for and retrieved a food pellet. There was no attempt to disorient the rat or prevent it from observing the refuge rotation in most cases; however, in cases when the rotations occurred with the rat in another room, there was no difference in the amount the preferred firing direction shifted compared to sessions when the rat was present during the rotation. Returning to the wrong doorway was counted as an error, and the angular displacement from the correct doorway was recorded. Errors were scored and the firing properties of HD cells were monitored during each session.

HD cell recordings in the food-carrying task provided interesting insights into how rats use HD cells for navigation. First, HD cells did not shift their preferred firing direction between the baseline cylinder sessions and the food-carrying sessions for the standard, blindfold, or dark conditions, but did shift during the Rotation condition. The amount of shift in the preferred firing direction in the rotation session usually corresponded to the amount the refuge position was moved from the training position (e.g., 90° or 180°), suggesting that under light conditions the HD cells were using the refuge as a landmark for orientation. Second, the rats rarely made errors during the food-carrying task while an HD cell was being recorded. On trials when the rat returned to the correct doorway, the cell's preferred firing direction usually did not shift, and was therefore consistent with the choice behavior of the rat (standard: 89%, blindfold: 100%, dark: 100%, rotation: 90%). However, during trials in which the rat made an error the cell's preferred firing direction usually did *not* shift to correspond to the behavioral choice; for example, if the animal returned to a doorway that was located 90° clockwise (CW) from the correct doorway, the preferred firing direction of the recorded HD cell did *not* shift 90° CW. Thus, the rat made a behavioral error, even though HD cell discharge remained constant compared to previous trials. The percentage of error trials in which the HD cell's preferred firing direction shifted a similar amount as the behavioral choice were as follows: standard: 14%, blindfold: 0%, dark: 0%, rotation: 17%. These results again demonstrate that the animal's behavioral response is not always coupled with the spatial information encoded by the HD cell network.

Some evidence of a relationship between the HD cell system and path integration was found, however, in the initial errors made following rotation of the refuge. As previously

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mentioned, the preferred firing direction of an HD cell would often rotate 180° if the refuge (and hence the starting point) was moved 180° opposite from the original training position, and in these cases the rat's choice behavior was similar to the other conditions (i.e., errors were rare). However, in a number of sessions the preferred firing direction did not shift completely with the amount of rotation of the refuge (i.e., underrotation). In these sessions, the rat made an error on the first one or two trials of the four-trial session, and in each case it made the behavioral error in a direction consistent with the amount of underrotation of the HD cell's preferred firing direction. Figure 11.3B shows one such example. In this case, the cell had an initial preferred firing direction of 300° and the rat's directional heading upon selecting a doorway was 225°, which was 75° CW from the cell's preferred firing direction. Following a 180° rotation of the refuge, the cell's preferred firing direction shifted by only 105°. However, the final directional heading of the rat as it approached the periphery to make a door choice *also* shifted by 105°. Thus, the final directional heading of the rat could be predicted by the shift in the cell's preferred firing direction.

This subset of results is consistent with the view that shifts in the preferred firing direction of the HD cell signal may initially guide a rat's choice behavior when it perceives a change in the surrounding environment, and if the preferred firing direction of the HD cell is misdirected, then the choice behavior is misdirected as well (figure 11.4). However, it was also found that, once the rat began making rotationally correct choices (i.e., choices oriented 180° relative to the initial doorway), the cell's preferred firing direction did not shift, and remained relatively stable throughout the four-trial session. One possibility is that, after the rat has selected the incorrect (and closed) doors, it started choosing other doors on subsequent trials, without a shift in its HD cell orientation.

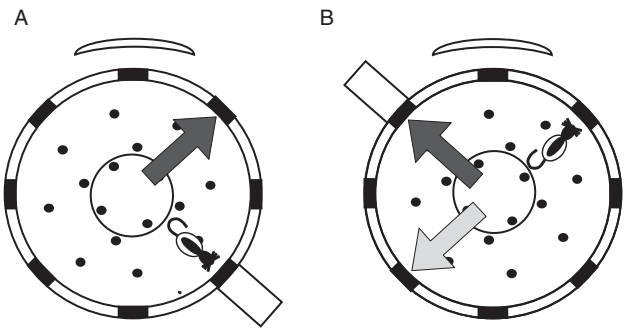


Figure 11.4
Relationship between the preferred firing direction and errant choice in a rotation session. The white cue curtain remained fixed between the initial standard session (A) and the subsequent rotation session (B), while the refuge was rotated 180° from the standard position. (A) During the standard session the preferred firing direction of the HD cell (dark gray arrow) was anchored ~90° relative to the behavioral choice of the rat (doorway to the refuge). (B) During the first trial of the 180° refuge rotation the preferred firing direction of the HD cell shifted by only 90° (dark gray arrow) instead of the full 180° (light gray arrow). The rat's behavior (as shown) appeared to correlate with the shift in the firing direction of its HD cells, and not the shift in the refuge location.

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It is worth considering the behavioral steps involved between the rat's perception of the cue on a rotation trial and the moment the rat makes a navigational choice. During rotation trials shortly after the rat emerged from the refuge, the preferred firing direction of the HD cell shifted some amount. If the amount of shift nearly matched the amount of refuge rotation (i.e., 180° for both), the rat did not make a behavioral error. If, however, the preferred firing direction underrotated (e.g., 90° cell rotation for a 180° refuge rotation), then the rat made one, or two, behavioral mistakes in the direction indicated by the information encoded by the HD cell, that is, to the place where the refuge should have been, as indicated by the underrotated HD cell, two doors away from the real refuge (figure 11.4). After the error(s) were made, there appeared to be a sudden realization that something was amiss, and on subsequent trials the rat made the correct choice. Nonetheless, this change in behavioral response did not affect the preferred firing direction of the recorded HD cell, which remained stable across all the rotation trials. It appears that the problem during the error trials was a combination of two processes: (1) the HD cell's preferred firing direction was not correctly anchored to the extra-maze cue curtain once the rat left the refuge (i.e., the HD cell was registering the cue as west instead of south), and (2) the HD cell's preferred firing direction had shifted from its original orientation in the refuge, therefore, the rat followed the errant HD cell system signal on its return trip. It is unlikely that the error was the result of the HD cell network incorrectly guiding the rat's behavior because the information provided by the HD cell network appeared to initially guide the navigational choice on the first couple of trials.

The Maintenance of a Sense of Direction across Environments by the HD Cell System

A second experiment was designed to test whether the HD cells system might allow the rat to use spatial information across two environments. This experiment (Dudchenko and Zinyuk, 2002) was based on the ability of rats to alternate directions on a T-maze. In this maze, a rat is typically given two consecutive runs from the base of the T, and there is a strong tendency for the animal to select different arms of the T on each run. Rats will do this even without training; this behavior is known as spontaneous alternation.

In an intriguing set of experiments, Douglas (1966) attempted to find out what cues the rats used to remember which arm of the T they had entered on their first run. To preclude the use of any olfactory or local cues, he gave rats their second run on a separate T-maze located in a different room. In doing this, Douglas made a striking observation: rats would alternate reliably when the two T-mazes were oriented in the same direction, but alternated at chance levels when the mazes were oriented in different directions relative to each other. This suggests that the rats were not simply making alternate egocentric responses (e.g., left on one maze, then right on the other), because this would be successful regardless of the second maze's orientation. Rather, Douglas concluded that the rats must be alternating arms based on a sense of direction that was carried across environments. This strategy

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would allow the rats to alternate when the mazes were parallel to one another, but would be insufficient when the mazes were in a different orientation.

To test whether the HD cell system could be the basis for this directional sense, Dudchenko and Zinyuk (2002) recorded from HD cells on T-mazes located in adjacent environments. The data shown in figure 11.5 are examples from one rat, and the results were consistent across subjects.

As is evident in figure 11.5A, when the rat was carried from a T-maze in one environment to a parallel T-maze in a second environment, the preferred direction of an HD cell was maintained. In figure 11.5B, however, it is clear the when the T-maze in the second room was rotated by 90°, the direction of the cell shifted by 90°. Thus, the HD cell was anchored to the maze and was not maintaining a sense of direction across environments. This result would be inconsistent with the hypothesis that the HD cell system underlies alternation across environments. However, alternation performance levels were low in both the parallel and perpendicular maze recording trials, so it is difficult to draw a firm conclusion about the relationship between HD cells and alternation across mazes.

The apparent anchoring of the HD cell system to the T-mazes raises an additional question: Why did the HD cells not carry their preferred directions across environments? In this study, no attempt was made to disorient the animals as they were carried from one T-maze to the other. Indeed, they were carried between these mazes in a shallow, clear plastic container. One possibility is that the reliable shifts in the cell's preferred direction to agree with the orientation of the second T-maze is an indication that familiar landmarks exert much more stimulus control over the HD cell system than does self-motion information (Dudchenko and Zinyuk, 2003).

HD Cells and the Tolman Sunburst Maze

In a recent experiment, Muir and Taube (2004) tested whether the information provided by the HD system was used to guide behavior in a navigational task requiring a cognitive mapping strategy. The firing properties of ADN and postsubicular (PoS) HD cells were recorded as rats performed a navigational task based on the classic "sunburst" maze study of Tolman et al. (1946). By training animals on a fixed maze configuration, then altering the maze by blocking the old route to the reward and introducing a new route, Tolman was able to demonstrate that rats could use a previously unavailable shortcut to reach a goal. This was seen as critical evidence to support the notion that animals possess an internal representation of space, or "cognitive map," that can be used by the animal while navigating. Muir and Taube adapted Tolman's maze by reducing the number of novel maze arms available from 18 to 8 to increase the angular difference between arms, and by removing the light from above the goal location which may have acted as a beacon in the original study.

Once trained on a maze configuration (e.g., figure 11.6A), animals ran three additional training trials along the elevated maze route in the same configuration to locate a water

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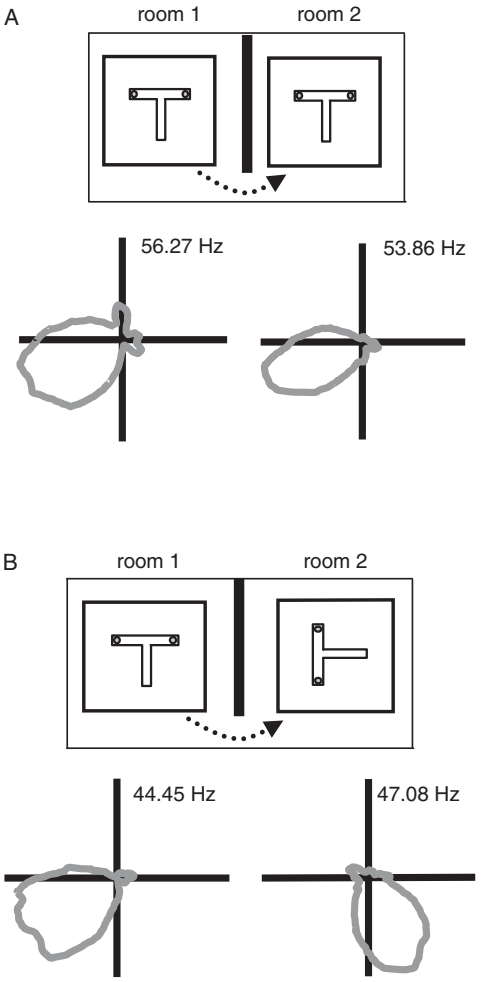


Figure 11.5
An anterior thalamic HD cell recorded across T-mazes in adjacent environments. This HD cell maintained a constant firing direction across environments when the T-mazes were parallel (A), but not when the mazes were perpendicular to one another (B) (Dudchenko and Zinyuk, 2002). The polar plots are in the absolute reference frame of the room.

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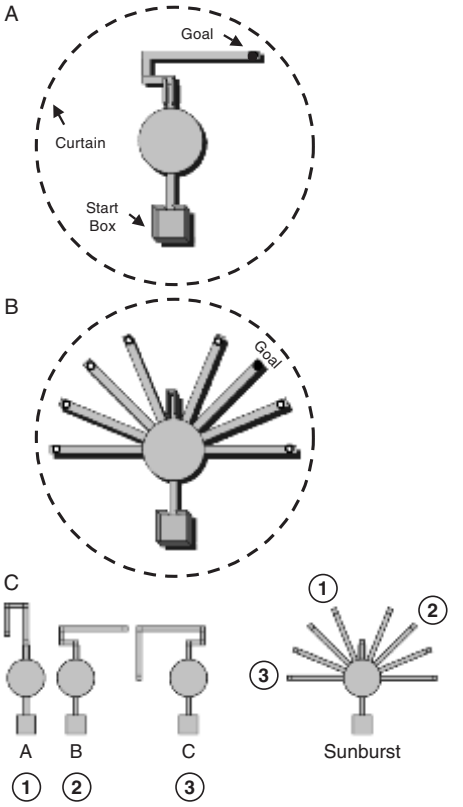


Figure 11.6
(A) An example of a maze configuration used during training trials. (B) The sunburst maze configuration. Note that the goal is in the same spatial location in the room as in A. (C) Each training maze used has a different path that is associated with a different correct shortcut arm (shortest path) on the sunburst trial, as shown on the numbered arms of the sunburst configuration on the right. (Reproduced from Muir and Taube, 2004, with permission.)

reward in a fixed location. The maze was enclosed within a black circular curtain throughout training and testing. Training trials began with an initial period during which animals were confined to the start box, followed by a period when the animal had access to the maze. Upon reaching the goal, the animal was returned to the start box and a lid placed over it for a delay. These three training trials were followed by a sunburst trial in which the normal route to the reward was blocked and eight novel routes introduced; one being a direct path (shortcut) to the reward's previous location (figure 11.6B). Animals that can utilize an accurate cognitive map of the environment should have been able to use the novel shortcut route to reach the reward. Once a sunburst trial was completed, the animal began training on a new maze configuration (figure 11.6C) with a different arm as the "correct" shortcut. For each configuration, data from the three training trials were com-

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bined and compared to data from the sunburst trial to determine the amount of shift in the cell's preferred firing direction.

The results showed that throughout the sunburst trial, the preferred direction of both ADN and PoS HD cells remained stable relative to the training trials and was not related to the accuracy of the behavioral choice(s) made by the animal. For example, when one animal was exposed to the sunburst maze for the first time and correctly selected the novel shortcut to the rewarded arm, the preferred firing direction of its HD cell did not shift, compared to the training trials on the first maze (figure 11.7A). Similarly, another animal that made seven errors before selecting the correct arm showed stable HD cell activity across trials on the maze, both compared to the training trials and to the sunburst trial (figure 11.7B). Overall, the preferred firing direction of HD cells on the sunburst maze was shifted, on average, by less than 1° (figure 11.7C) from that seen during the training trials, while animals averaged 5.5 errors (range: 0–25) per sunburst trial.

Additional probe trials conducted on cells by rotating the maze 180° relative to the room showed that the firing of the HD cells was related to the orientation of the maze and not the room, suggesting that the animals were using the maze as an orienting cue. If animals were indeed using the maze as a cue to orient by, and given that the central platform of the maze and the start box, in particular, remained constant from the training trials to the sunburst trials, it is not surprising that the HD cell activity also remained unaltered from training to sunburst trials. Although the animals may have lacked sufficient experience with the novel "sunburst" component of the task to perform well, one animal that experienced the sunburst maze nine times consistently showed HD cell activity unrelated to its behavioral choice across all nine sunburst trials. The results clearly show that HD cell activity, at least in the PoS and ADN, remained robust throughout all phases of the task, independently of whether the animal performed well or poorly. This finding stands in contrast to the results reported by Mizumori and Williams (1993), who suggested that HD cells in the LDN developed a more robust directional signal as an animal learned the task. However, these data are consistent with those of Golob et al. (2001): the animal may have reliably encoded directional information about its orientation upon entry into the environment, but this information may not necessarily have been used to guide the animal's navigation to a goal.

A Digression: the Head Direction Cell System and Disorientation

HD Cells and Landmark Learning

We have seen that in a nonpolarized apparatus such as the radial arm maze, the same extramaze cue—a white curtain—can exert stimulus control over the rats' spatial behavior and the preferred firing direction of their HD cells. In this type of situation, how do landmarks in the environment come to anchor the preferred firing direction of HD cells? One hypothesis is that this stimulus control is the result of a learned association between visual

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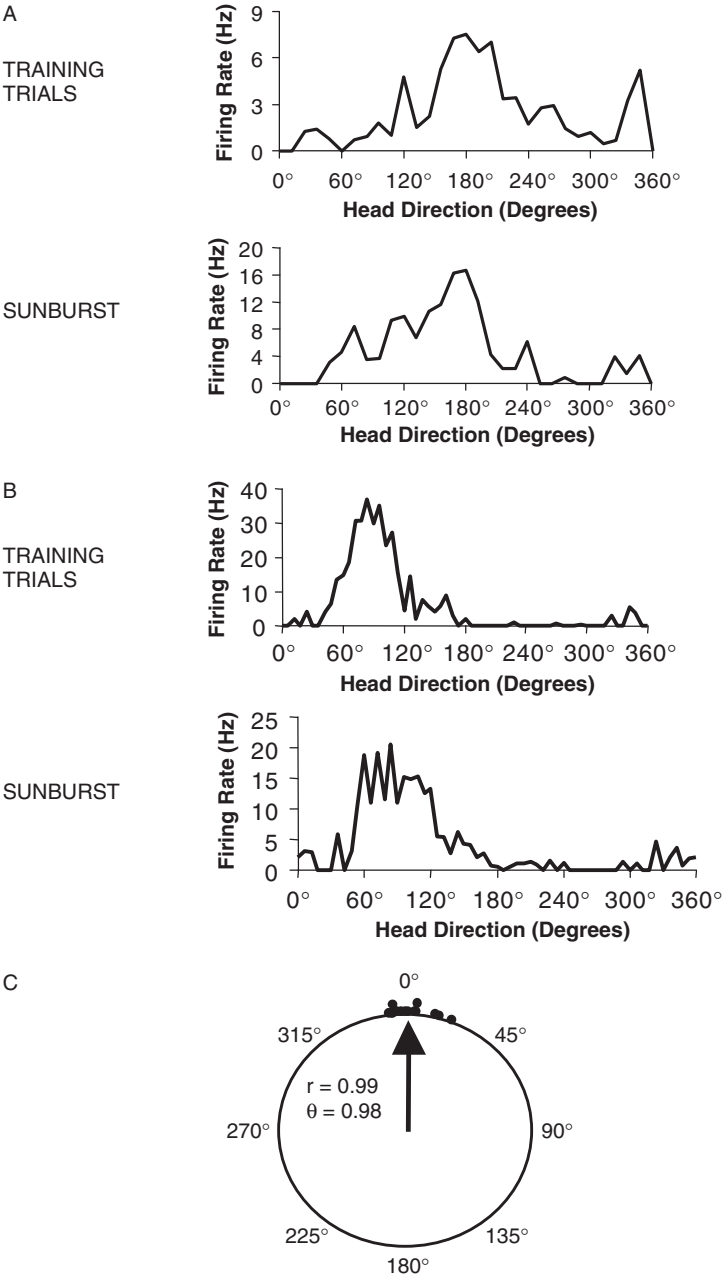


Figure 11.7
(A) On this sunburst trial, the animal chose the correct shortcut arm on the first choice, and the preferred firing direction (although less clear due to the low firing rate and short sampling times for this cell) remains stable from the training trials to the sunburst trial. (B) The preferred firing direction of this ADN HD cell was unaltered from the training trials to the sunburst trial, during which the animal made seven errors before choosing the correct shortcut arm. This HD cell stability was consistent for the duration the animal was on the maze. (C) Distribution of shifts in the preferred firing direction of HD cells from training trials (=0°) to the sunburst trial. The shift for each given cell is shown as a filled circle on the outside of the unit circle with degrees marked. As is apparent, the average amount of shift was negligible (~1°), with very little angular deviation observed. r, mean vector length; θ, mean phase angle. (Reproduced from Muir and Taube, 2004, with permission.)

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landmarks, and a sense of orientation derived from self-motion cues (McNaughton et al., 1991, 1996).

Empirical support for this view came from an experiment conducted by Knierim et al. (1995; also see chapter 8 by Knierim). They sought to break the link between the rat's sense of orientation and a cue card in a cylinder by repeatedly disorienting the rat before it was placed in the cylinder and after it was removed. Disorientation took the form of gentle rotation of the rat in an opaque container as the experimenter approached the cylinder. The authors observed that HD cells and hippocampal place cells in rats that had been repeatedly disoriented were less likely to be anchored to the cue card in the cylinder. This weakening of the cue control appeared to persist even when the disoriented rats were subsequently trained under nondisorientation conditions.

As Knierim et al. suggest, this result is slightly counterintuitive: one might expect that a disoriented rat would be more, not less, likely to derive its orientation from a stable visual landmark. Indeed, in other studies (using a different strain of rats), disorientation prior to recording appeared to have less effect on the stimulus control of a visual landmark over the preferred firing direction of HD cells and the place fields of hippocampal place cells (Taube, 1995; Dudchenko et al., 1997b; Golob and Taube, 1997). However, the results of Knierim et al. would imply that disoriented rats would have difficulty learning to use extra-maze landmarks to guide their spatial behavior.

To test this, Dudchenko et al. (1997a) trained disoriented and nondisoriented rats on a landmark-based, eight-arm radial maze task. The maze was curtained off from the remainder of the environment, and a white curtain served as the sole extra-maze landmark available to the rats. A single maze arm contained a water reward, and the rat's task was to reliably select this arm when placed on the center of the maze. Although seven out of eight animals in one of the nondisoriented groups learned the task, none of the eight disoriented animals learned it.

This behavioral result supports the view that landmarks are learned by their association with a stable internal sense of orientation. One would predict, then, that rats should be impaired on any type of landmark-based task when they are first disoriented prior to the task. However, this prediction appears not to be the case. Both in the Dudchenko et al. study, and a study done at the same time by Martin et al. (1997), disoriented rats readily learned to use a landmark to solve a Morris water-maze task. This interaction between disorientation and type of spatial task—aversive versus appetitive—has also been replicated in landmark navigation tasks in a circular arena (Gibson et al., 2001) and a rectangular pool (Golob and Taube, 2002). In addition, disoriented rats can learn to solve an aversive plus maze task (Hynes et al., 2000).

Interim Summary

The stimulus control of visual landmarks over HD cell preferred firing appears to be learned, and learned relatively quickly, within a minute or two (see chapter 3 by Taube).

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Disruption of this association by disorientation weakens this stimulus control in some situations, and results in a clear impairment in the use of landmarks to guide spatial behavior in appetitive spatial tasks. Why landmarks can be learned by disoriented animals when they are placed in aversively motivated tasks, remains an open question. However, one implication of this dissociation is that, in the disoriented rat, neural systems in addition to the HD cell and place cell systems can be used to guide spatial behavior when escape from an aversive condition is required. In support of this notion, recent work in *Drosophila* shows that there are two independent neurotransmitter systems for learning appetitive (octopamine) and aversive (dopamine) olfactory tasks (Schwaerzel et al., 2003).

Disorientation and HD Cell Activity The issue of behavioral disorientation raises the question: What are the neural correlates for disorientation? How do HD cells respond when an animal is behaviorally (or perceptually) disoriented? Do HD cells lose their direction-specific firing? If so, do the cells become completely quiescent (i.e., total absence of discharge) or is there a tonic rate of firing that is elevated above background firing levels? Understanding the neural correlates of disorientation is also important in the context of the three-dimensional locomotion experiments described in chapter 3 by Taube. These experiments showed that when the rat locomoted on the ceiling in 1 g, or on the wall or ceiling in 0 g, there was a loss of direction-specific firing (Calton et al., 2000; Taube et al., 2004). Were these animals "perceptually" disoriented at the same time? A better understanding of the neural correlates underlying spatial disorientation would help to resolve this issue.

Three forms of spatial disorientation have been identified in humans (Gillingham and Previc, 1996). Type I spatial disorientation involves a misperception of one's orientation and is unrecognized by the observer. Assuming the activity of HD cells reflects the sense of direction of the animal, direction-specific firing of HD cells would presumably be maintained in this situation, albeit at an incorrect orientation with respect to the environment. Type II spatial disorientation entails a conscious recognition by the subject that he or she is disoriented and attempts are then made to become oriented by using any available information. Type III spatial disorientation occurs when the subject becomes so disoriented that he or she is incapacitated. This type of spatial disorientation can occur when the subject is experiencing rapid and continual rotations that lead to confusion, or when the subject experiences severe motion sickness or oscillopsia that makes it difficult to override these compelling conditions. In contrast to type I spatial disorientation, types II and III spatial disorientation would be expected to lead to a disruption of normal HD cell discharge. Indeed, type I spatial disorientation would more appropriately be labelled a "misorientation" rather than disorientation. The type of disorientation that is relevant to the questions posed here is most closely associated with types II or III.

Knierim et al. (1998) reported that HD cell firing occasionally (2 out of 10 cases) became "erratic, firing with no directional specificity" when rats were spun continuously

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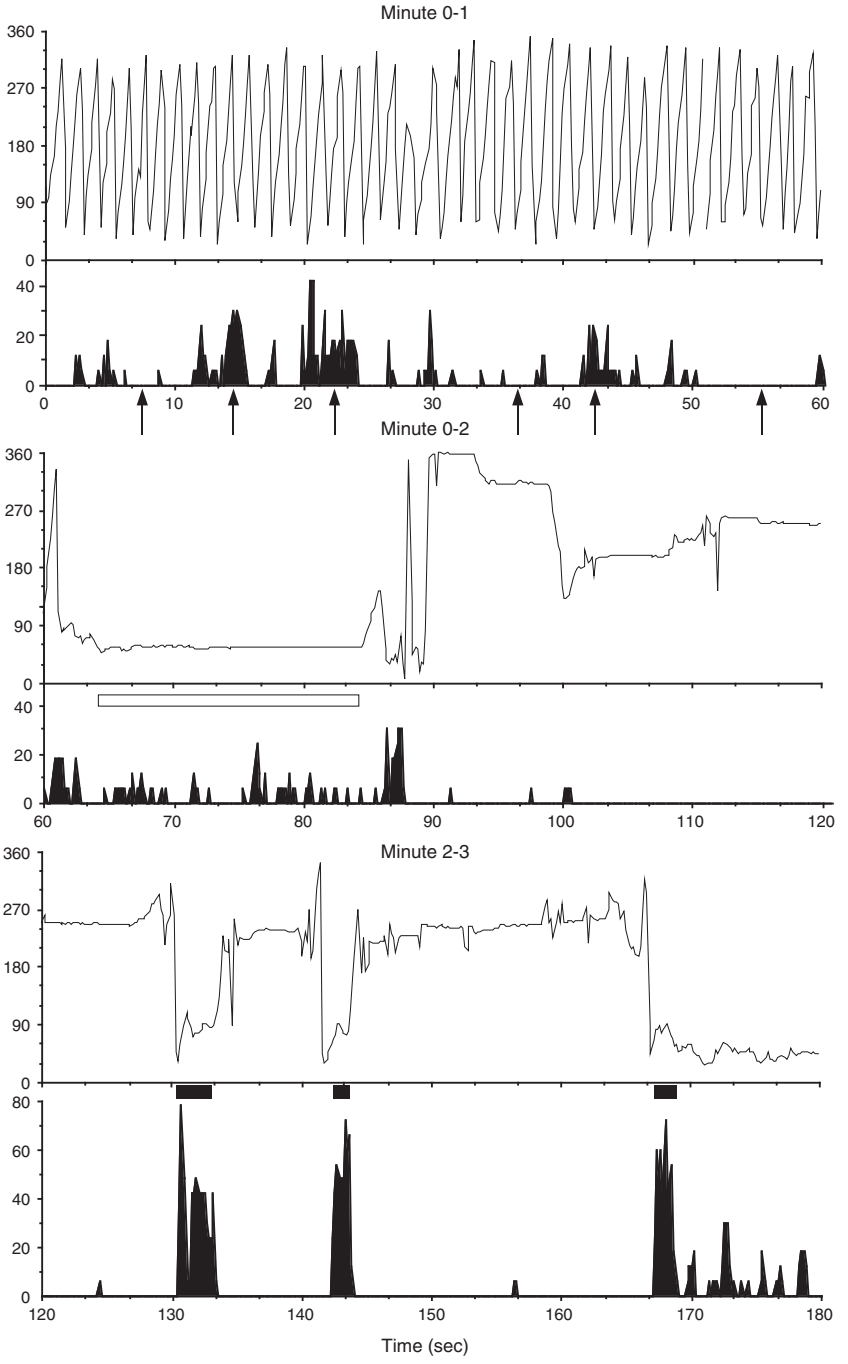
in one direction (unidirectional) on a rotating pedestal in the dark. While continuing to spin the rats, cell firing was also described as erratic in 6 out of 10 cases when the lights were turned back on and the animal could view a prominent landmark within the room. Erratic cell firing was also occasionally observed (5 out of 16 cases) after some time when rats were spun irregularly back-and-forth (bidirectional) for several minutes. The erratic discharge observed in these cells could best be described as cell firing that occurred in many different directions, although detailed analyses over small time intervals was not conducted. It should be noted that these experiments were conducted on Fischer-344 rats, animals that are thought to have poor vision (Munn, 1950). This fact may have contributed to the poor control the landmark exerted over cell firing once the lights were turned back on.

To address this issue more extensively, Steven and Taube (2002) spun Long-Evans rats back and forth on a turntable while recording HD cell responses. Long-Evans rats have better visual capacities than Fischer-344 rats (Munn, 1950). In addition, cell firing was analyzed with a greater temporal resolution than in the Knierim et al. study. The animals were either blindfolded and spun in the dark, or were spun without blindfolds in the light with a prominent visual cue that has been shown to exert control over HD cell preferred firing directions. Spinning speeds were either slow (approx. 90°/s) or fast (approx. 240° s). The results were quite clear. When rats were spun under blindfold conditions at either slow or fast speeds, HD cells quickly lost their direction-specific firing within one to two revolutions and cell activity varied between states of increased tonic firing at all head directions to total quiescence (figure 11.8). In contrast, when rats were spun without blindfolds in the light, direction-specific firing was maintained and the cells' preferred directions remained in alignment with the prominent visual cue.

These results raise the interesting chicken and egg question: Which comes first? Do the HD cells lose their direction-specific firing and therefore the animal perceives that it is disoriented? Or, does the animal become disoriented as a result of the rotary stimulation, and therefore HD cells lose their direction-specific firing? Of course, the answer to this question is not clear from these experiments, but one would like to think that the neural circuitry responsible for detecting angular velocity was not capable of following the constant irregular rotary stimulation. This incapacity then led to erroneous signals being passed on to the HD cell network, which in turn resulted in the loss of direction-specific discharge. The readout from the HD cell network thus did not provide accurate information concerning directional heading, and the animal would subsequently perceive itself to be disoriented. Alternatively, it is also possible that circuitry outside the HD cell network might have been disrupted by the rotary stimulation, and this disruption in turn would have led to erroneous signals being passed onto the HD cell network. To distinguish these possibilities, future experiments might test whether it is possible to dissociate HD cell discharge from the perception of disorientation. For example, can conditions be created where either (1) HD cell activity is disrupted without the animal perceiving that it is disoriented?

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or (2) where the animal perceives itself to be disoriented despite normal HD cell discharge? If it proves impossible to create either of these conditions, then this result would support the notion that the loss of direction-specific firing is the neural correlate of perceived disorientation.

Secondary Correlates of HD Cells on the Radial-Arm Maze

A number of previous studies have shown that hippocampal place cells contain additional, nonspatial neuronal correlates when tested under the right conditions (e.g., West et al., 1981; Berger et al., 1983; Breese et al., 1989; Wiener et al., 1989; Young et al., 1994; Wood et al., 2000). For example, Wiener et al. (1989) demonstrated that place cell firing can be correlated to more than one behavioral or spatial event, depending on the type of behavioral task the animal is performing. Consequently, one might ask whether HD cells show nonspatial correlates when also tested under the right conditions. The limited work that has addressed this issue has not shown any evidence for such activity, at least in ADN HD cells. Dudchenko and Taube (1997) recorded eight HD cells on the radial-arm maze task and explicitly looked for activity that was modulated by the rat's behavior other than directional heading. Specifically, they looked for evidence of goal-approach or reward-consumption correlates in HD cell activity as the rat travelled down a reinforced arm and consumed a water reward at the end of the arm. The activity that was observed was always related to the rat's head direction, and this activity did not appear to be modulated by the approach to or consummation of the reward. Thus, in contrast to place cells, there was no evidence that ADN HD cell firing was modulated by the rat's behavior other than directional heading in this spatial reference memory task.

What Does the Head Direction Cell System Do?

The clear, robust activity of HD cells would appear to make this neural system an ideal substrate for spatial cognition. However, attempts to link the behavior of HD cells to the

Figure 11.8
Firing rate and head direction by time plot of a HD cell for a 3-minute session. During the first minute (top graph), a blindfolded rat was spun clockwise continuously on a turntable at approximately 270°/s. Direction-specific firing was lost, and cell firing could be characterized as either silent (black-filled arrows) or increased tonic, background firing (open arrows). After 1 minute the platform rotation was stopped and the rat was free to move around for 2 minutes to retrieve small pellets that were randomly dropped onto the floor of the platform (middle and bottom graphs). Direction-specific firing returned quickly with a preferred firing direction at 78°. Note that when the rat's head was pointing in the preferred firing direction, firing rate peaked at about 80 spikes/s (black-filled bars, bottom graph). When the rat maintained a head direction for about 16 sec that pointed about 20° clockwise of the cell's preferred firing direction (open bar, middle graph), the cell fired constantly at a rate below the peak firing rate. Firing rates during the first minute when the rat was spun never approached the level of peak firing during nonrotation conditions, although the cell occasionally fired at a relatively high rate for several seconds before going through subsequent silent periods.

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rat's spatial behavior have been only partially successful. Although a correlation between HD cells and behavior was observed by Dudchenko and Taube (1997), and to an extent by Frohardt et al. (2002), studies by Golob et al. (2001), and Muir and Taube (2004) show a clear lack of correlation. Importantly, a similar lack of correlation has also been seen in hippocampal place cells (Jeffery et al., 2003).

What, then, does the head direction system do? It seems likely that the HD cell system, like the place cell system, can use different types of cues to orient relative to an environment. In environments that do not possess a polarizing layout, such as the cylinder or radial maze, a polarizing cue (such as a cue card) will control the direction in which HD cells fire. If such a cue is not available or is ambiguous, as in the blindfold and dark trials on the circular maze of Frohardt et al., HD cells may anchor themselves to the point from which the rat enters the environment. However, in environments such as a square, rectangle, T-maze, or sunburst maze, the shape of the environment itself is a strong cue that can "anchor" the HD cell system. Thus, the likelihood that the HD cell system and the animal's spatial behavior are related depends on the aspect of the environment to which the spatial behavior is oriented. If, for example, a rat is trained to find a goal location that is defined by a spatial association with a cue card, the likelihood that this cue card will also anchor the rat's HD cell system depends on the shape of the environment. HD cells may automatically orient to particular aspects of the environment, such as a corner; rats, on the other hand, may be trained to make spatial associations with other cues in an environment.

A stronger test of the capacity of the HD cell system to guide behavior would require that the rat make a spatial response based on this system. As has been done with hippocampal place cells (O'Keefe and Speakman, 1987), this may be tested by assessing rats in an environment that lacks directional landmarks, and observing that the behavior can be predicted by the direction in which a given HD cell fires. The circular maze of Frohardt et al. is an attempt to do this, although the rats' HD cell systems appeared to anchor themselves to the start locations. Tests in which the HD cell system shifts from trial to trial, and in which a corresponding shift in the animal's spatial behavior is observed, would provide support for a role of this system in guiding spatial behavior.

Note

1. The relative influence of landmarks versus the geometry of the environment in controlling spatial behavior may present a challenge to traditional learning theory. Recent results by Pearce et al. (2001) and Hayward et al. (2003) suggest that the shape of the environment is not overshadowed or blocked by landmarks.

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References

Barnes CA (1979) Memory deficits associated with senescence: A neurophysiological and behavioral study in the rat. J Comp Physiol Psychol 93: 74–104.

Berger TW, Rinaldi PC, Weisz DJ, Thompson RF (1983) Single-unit analysis of different hippocampal cell types during classical conditioning of rabbit nictitating membrane response. J Neurophys 50: 1197–1219.

Breese CR, Hampson RE, Deadwyler SA (1989) Hippocampal place cells: Stereotypy and plasticity. J Neurosci 9: 1097–1111.

Calton JL, Tullman ML, Taube JS (2000) Head direction cell activity in the anterodorsal thalamus during upside-down locomotion. Soc Neurosci Abstr 26: 983.

Cheng K (1986) A purely geometric module in the rat's spatial representation. Cogn 23: 149–178.

Douglas RJ (1966) Cues for spontaneous alternation. J Comp Phys Psychol 62: 171–183.

Dudchenko PA, Davidson M (2002) Rats use a sense of direction to alternate on T-mazes located in adjacent rooms. Animal Cogn 5: 115–118.

Dudchenko PA, Goodridge JP, Seiterle DA, Taube JS (1997a) Effects of repeated disorientation on the acquisition of spatial tasks in rats: dissociation between the appetitive radial arm maze and aversive water maze. J Exp Psychol: Anim Behav Proc 23: 194–210.

Dudchenko PA, Goodridge JP, Taube JS (1997b) The effects of disorientation on visual landmark control of head direction cell orientation. Exp Brain Res 115: 375–380.

Dudchenko P, Taube JS (1997) Correlation between head-direction single unit activity and spatial behavior on a radial arm maze. Behav Neurosci 111: 3–19.

Dudchenko PA, Zinyuk LE (2002) Head direction cell orientation after cue card separation, and after transport between two mazes. Soc Neurosci Abstr 28: 584.5**.**

Dudchenko PA, Zinyuk LE (2003) Do head direction cells rely more on landmarks than path integration when a rat walks between environments? Soc Neurosci Abstr 29: 91.2.

Etienne AS, Maurer R, Saucy F (1988) Limitations in the assessment of path dependent information. Behaviour 106: 81–111.

Frohardt RJ, Marcroft JL, Taube JS (2001) Lesions of the anterior thalamus impair path integration performance on a food-carrying task. Soc Neurosci Abstr 27: 315.11.

Frohardt RJ, Marcroft JL, Taube JS (2002) Do head direction cells guide spatial navigation in rats? An electrophysiological investigation in a path integration task. Soc Neurosci Abstr 28: 584.1.

Gibson BM, Shettleworth SJ, McDonald RJ (2001) Finding a goal on dry land and in the water: differential effects of disorientation on spatial learning. Behav Brain Res 123: 103–111.

Gillingham KK, Previc FH (1996) Spatial orientation in flight. In R Dehard ed., Fundamentals of Aerospace Medicine. 2nd ed. Baltimore: Williams and Wilkins, pp. 309–397.

Golob EJ, Taube JS (1997) Head direction cells and episodic spatial information in rats without a hippocampus. Proc Nat Acad Sci USA 94: 7645–7650.

Golob EJ, Taube JS (2002) Differences between appetitive and aversive reinforcement on reorientation in a spatial working memory task. Behav Brain Res 136: 309–316.

Golob EJ, Stackman RW, Wong AC, Taube JS (2001) On the behavioral significance of head direction cells: neural and behavioral dynamics during spatial memory task. Behav Neurosci 115: 285–304.

Hayward A, McGregor A, Good MA, Pearce JM (2003) Absence of overshadowing and blocking between landmarks and the geometric cues provided by the shape of the test arena. Quart J Exp Psychol 56B: 114–126.

Hermer L, Spelke ES (1994) A geometric process for spatial reorientation in young children. Nature 370: 57–59.

Hynes CA, Martin GW, Harley CW, Huxter JR, Evans JH (2000) Multiple points of entry into a circular enclosure prevent place learning despite normal vestibular orientation and cue arrays: evidence for map resetting. J Exp Psychol: Anim Behav Proc 26: 64–73.

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Jeffery KJ, Gilbert A, Burton S, Studwick A (2003) Preserved performance in a hippocampal-dependent spatial task despite complete place cell remapping. Hippocampus 13: 175–189.

Knierim JJ, Kudrimoti HS, McNaughton BL (1995) Place cells, head direction cells, and the learning of landmark stability. J Neurosci 15: 1648–1659.

Knierim JJ, Kudrimoti HS, McNaughton BL (1998) Interactions between idiothetic cues and external landmarks in the control of place cells and head direction cells. J Neurophys 80: 425–446.

Lashley KS (1929) Brain mechanisms and intelligence. Chicago: University of Chicago Press.

Margules J, Gallistel CR (1988) Heading in the rat: Determination by environmental shape. Anim Learn Behav 16(4): 404–410.

Martin GW, Harley CW, Smith AR, Hoyles SE, Hynes CA (1997) Spatial disorientation blocks reliable goal location on a plus maze but does not prevent goal location in the Morris maze. J Exp Psychol: Anim Behav Proc 23: 183–193.

McNaughton BL, Chen LL, Markus EJ (1991) "Dead Reckoning," landmark learning, and the sense of direction; a neurophysiological and computational hypothesis. J Cogn Neurosci 3: 190–201.

McNaughton BL, Barnes CA, Gerrard JL, Gothard K, Jung MW, Knierim JJ, Kudrimoti H, Qin Y, Skaggs WE, Suster M, Weaver KL (1996) Deciphering the hippocampal polyglot: the hippocampus as a path integration system. J Exp Biol 199: 173–185.

Mizumori SJY, Williams JD (1993) Directionally selective mnemonic properties of neurons in the lateral dorsal nucleus of the thalamus of rats. J Neurosci 13: 4015–4028.

Muir GM, Taube JS (2004) Head direction cell activity and behavior in a navigation task requiring a cognitive mapping strategy. Behav Brain Research, in press.

Muir GM, Taube JS (2002) The neural correlates of navigation: do head direction and place cells guide spatial behavior? Behav Cogn Neurosci Rev 1: 297–317.

Munn NL (1950) Handbook of Psychological Research on the Rat. Boston, MA: Houghton Mifflin, pp. 155–156.

O'Keefe JA, Speakman A (1987) Single unit activity in the rat hippocampus during a spatial memory task. Exp Brain Res 68: 1–27.

Pearce JM, Ward-Robinson L, Good M, Fussell C, Aydin A (2001) Influence of a beacon on spatial learning based on the shape of the test environment. J Exp Psychol: Anim Behav Procs 27: 329–344.

Small WS (1901) Experimental study of the mental processes of the rat. II. Am J Psychol 12: 206–239.

Schwaerzel M, Monastirioti M, Scholz H, Friggi-Grelin F, Birman S, Heisenberg M (2003) Dopamine and octopamine differentiate between aversive and appetitive olfactory memories in *Drosophila*. J Neurosci 23: 10495–10502.

Steven MS, Taube JS (2002) Head direction cell discharge during periods of disorientation. Soc Neurosci Abstr 28: 584.3.

Taube JS (1995) Head direction cells recorded in the anterior thalamic nuclei of freely moving rats. J Neurosci 15: 70–78.

Taube JS, Muller RU, Ranck, JB Jr. (1990) Head direction cells recorded from the postsubiculum in freely moving rats. II. Effects of environmental manipulations. J Neurosci 10: 436–447.

Taube JS, Stackman RW, Calton JL, Oman CM (2004) Rat head direction cell responses in 0-G parabolic flight. J Neurophys, in press.

Tolman EC, Ritchie BF, Kalish D (1946) Studies in spatial learning. I. Orientation and the short-cut. J Exp Psychol 36: 13–24.

Watson JB (1907) Kinaesthetic and organic sensations: their role in the reactions of the white rat to the maze. Psych Rev (Monog Supp) Vol. 8, no. 2

West MO, Christian E, Robinson JH, Deadwyler SA (1981) Dentate granule discharge during conditioning. Exp Brain Res 44: 287–294.

Whishaw IQ, Tomie JA (1997) Piloting and dead reckoning dissociated by fimbria-fornix lesions in a rat food-carrying task. Behav Brain Res 89: 87–97.

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Whishaw IQ, Hines DJ, Wallace DG (2001) Dead reckoning (path integration) requires the hippocampal formation: evidence from spontaneous exploration and spatial learning tasks in light (allothetic) and dark (idiothetic) tests. Behav Brain Res 17: 49–69.

Wiener SI, Paul CA, Eichenbaum H (1989) Spatial and behavioral correlates of hippocampal neuronal activity. J Neurosci 9: 2737–2763.

Wood ER, Dudchenko PA, Robitsek RJ, Eichenbaum H (2000) Hippocampal neurons encode information about different types of memory episodes occurring in the same location. Neuron 27: 623–633.

Young BJ, Fox GD, Eichenbaum H (1994) Correlates of hippocampal complex-spike cell activity in rats performing a nonspatial radial maze task. J Neurosci 14: 6553–6563.

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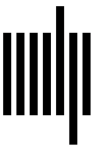
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